# Yiyi Cai

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## Education

2026– **Stanford University**, Stanford, CA
Doctor of Philosophy, Computer Science

2025-26 **University of Cambridge**, Cambridge, UK
Master of Philosophy, Advanced Computer Science, with distinction

2021-25 **California Institute of Technology**, Pasadena, CA
Bachelor of Science, Electrical Engineering, GPA 4.1/4.3

## Papers (* indicates alphabetical ordering)

- **Yiyi Cai***, Yongtao Zhan*, Alexander Zlokapa*. Fast mixing of the SYK model at high temperatures. arXiv:2610.05953.
- Jinge Bao*, **Yiyi Cai***, Zhong-Xia Shang*. Optimal tolerant testing of Lindbladian dissipation. arXiv:2609.38160.
- Akshar Ramkumar, **Yiyi Cai**, Yu Tong and Jiaqing Jiang. High-temperature fermionic Gibbs states are mixtures of Gaussian States. *Physics Review Letters*, vol. 137, no. 2, p.020601, 2026. **QIP 2026**.
- **Yiyi Cai**. Optimal detection of dissipation in Lindbladian dynamics. arXiv:2603.17736.
- **Yiyi Cai**. Improving parameters of asymptotically good quantum LDPC codes via stronger product expansion. Senior Thesis (Major).
- **Yiyi Cai**, Yu Tong, and John Preskill. Stochastic error cancellation in analog quantum simulation. *In 19th Conference on the Theory of Quantum Computation, Communication and Cryptography*. **Outstanding Paper Prize at TQC 2024.**
- Elena Scire, Lee Bennett, Judy Adler, Sergio Fajardo-Acosta, Robert Fogg, Elise Furlan, Jacob Llamas, Peter Bahariance, **Yiyi Cai**, Trinity Chung, Garni Gharibian, Emily Hu, Julianna Jin, Matteo Kimura, Aaron Lee, Michael Mansour, Mahak Mathur, Andy Sun, Jasmine Terrones, Jingchao Zhong, and Bethany Ehlmann. Lunar trailblazer ground system development. *Proceeding SPIE 13098, Observatory Operations: Strategies, Processes, and Systems X*, 2024.

## Awards & Honors

2026 National Science Foundation Graduate Student Fellowship
2026 Stanford Graduate Fellowship
2025 Gates Cambridge Scholarship
2025 National Defense Science and Engineering Graduate Fellowship (declined)
2024 Outstanding Paper Prize at 2024 TQC

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| 2023 | Mellon Mays Undergraduate Fellowship |
| 2023 | Arthur R. Adams Summer Undergraduate Research Fellowship |
| 2022 | Doris Everhart Quantum Summer Undergraduate Research Fellowship |

## Research Experience

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| 2025 - 2026 | Student Researcher, University of Cambridge |
| 2023 - 2025 | Student Researcher, Institute of Quantum Information and Matter, California Institute of Technology |
| 2024 | Quantum Computing Research Analyst, Global Technology Applied Research, JP-Morgan Chase & Co. |
| 2022 | Student Researcher, Quantum Machine Learning for High Energy Physics, California Institute of Technology |
| 2021 - 2023 | Missions Operations & Ground Data Science Intern, Lunar Trailblazer Mission, NASA & California Institute of Technology |

## Talks & Seminars

- ◇ University of Waterloo Institute for Quantum Computing Math and CS seminar; June 2026
- ◇ Contributed talk at QIP; Jan. 2026
- ◇ Academia Sinica seminar; Aug. 2025
- ◇ Contributed talk at TQC; Sep. 2024
- ◇ Caltech Summer Undergraduate Research Fellowship Seminar; Aug. 2023
- ◇ Caltech Summer Undergraduate Research Fellowship Seminar; Oct. 2022

## Teaching

### The Coding School

Teaching Assistant & Curriculum Developer

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| Summer 2025 | Quantum Summer Institute |

### California Institute of Technology

Teaching Assistant

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|---|---|
| Spring 2025 | IDS 157 Statistical Inference |
| Fall 2025 | EE 111 Signal-Processing Systems and Transforms |
| Spring 2024 | EE 10b Embedded Systems |
| Winter 2024 | EE 10a Digital Logic |
| Fall 2024 | EE 55 Mathematics of Electrical Engineering |

## Academic Service

**Journal Reviews**

PRX, PRX Quantum

**Conference Reviews**

QIP 2027, FOCS 2026, ICALP 2026, ITCS 2026, QIP 2026

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Last updated: October 14, 2026